



Software Development and Project Management Milestone #1 - Progress Report

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Selected Topic

Customer Complaint Monitoring

Declaration of the use of Generative AI.

Generative AI was used to support implementing sentiment analysis models and implementing processes in the code. Earlier part of report is written by human & AI. Later part of report is human-written.

The contents and codebase are reviewed by team members.

Product Overview

1. Product Description

Customer Complaint Monitoring is a sentiment-analysis system that identifies negative customer tweets.

By supporting companies with this system, helping detection of complaints and the feedback trend coming from customers, we allow CS team and related team to respond efficiently and effectively.

And ultimately help team create better solution by quickly receiving user feedback.

This system aims to support companies as an efficient tooling to support in analysis of the customer complaint.

2. Target Customers, Target Users and Market Segment

Target Customers

The target customers are medium-to-large companies that receive a high volume of customer comments through social-media platforms.

We anticipate the possible customers include airlines, delivery companies, online shopping platforms, banks and telecommunications companies.

Target Users

The target users are customer-support agents, social-media monitoring teams and customer-experience managers.

Market Segment

The initial market segment consists of companies that receive English-language customer tweets and need an automated method to prioritize possible complaints.

3. Stakeholder Requirements

Customer-support agents need to identify genuine customer complaints quickly. Missing a real complaint may damage customer satisfaction, but too many false complaint alerts may increase the workload of support agents.

Therefore, the system should prioritize detecting negative tweets while maintaining acceptable prediction precision.

4. Ready-to-Ship Criteria

A candidate model will be considered ready to ship if the f1 score and the accuracy remains as a meaningful reference for target users (cs agents, etc) to support their work. We set it as 70% initially.

Among the negative recall and accuracy, negative recall would be prioritized in evaluation because the system should avoid missing genuine customer complaints.

The precision requirement is included to prevent excessive false complaint alerts for customer-support agents.

Data Introduction & Inspection

1. TweetEval Dataset

This project uses the sentiment subset of the TweetEval benchmark.

The dataset contains English tweets with ground-truth sentiment labels and provides fixed training, validation and test splits.

The three sentiment classes are negative, neutral and positive.

Dataset source: https://huggingface.co/datasets/cardiffnlp/tweet_eval

2. Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand the structure, class distribution and quality of the TweetEval sentiment dataset.

We examined dataset sizes, label mapping, sentiment distribution, missing values, duplicate tweets, sample tweets and tweet lengths.

We examined the dataset through multiple steps following:

1. **Dataset Split Sizes:** This examines the number of tweets in the train, validation, and test sets respectively.

Split	Number of Tweets
Train	45615
Test	12284
Validation	2000

1. **Sentiment Label Mapping:** This figure out the meaning of numeric label in the data and convert it into a table.

Numeric Label	Meaning
0	negative
1	neutral
2	positive

3. **Sentiment Class Distribution:** To identify possible imbalance in classes, we counted the positive, neutral, and negative data in each data split.



4. **Data-Quality Checks:** We checked each data split for missing text, empty tweets, missing labels and duplicate tweets.

	Split	Missing Text	Empty Text	Missing Labels	Duplicate Tweets
0	Train	0	0	0	29
1	Validation	0	0	0	0
2	Test	0	0	0	0

Examples of duplicate tweets:

	text	label	sentiment
12396	"Charlie Rose with Bill Joy; Daniel Barenboim ...	1	Neutral
22568	"Charlie Rose with Bill Joy; Daniel Barenboim ...	1	Neutral
338	"Charlie Rose with Desmond Tutu; Bill Joy (Oct...	1	Neutral
19532	"Charlie Rose with Desmond Tutu; Bill Joy (Oct...	1	Neutral

Examples of duplicate text with conflicting labels (count: 3)

	text	label	sentiment
16929	An emotionally intelligent organisation throug...	1	Neutral
18334	An emotionally intelligent organisation throug...	2	Positive
43865	Buses to replace train services on Sunday as f...	1	Neutral
16293	Buses to replace train services on Sunday as f...	2	Positive
45079	Not making it to the main debate stage may not...	0	Negative
28331	Not making it to the main debate stage may not...	1	Neutral

5. **Tweet-Length Analysis:** Tweet length may influence model performance. We examined both the number of characters and the number of words.

	Character Count	Word Count
count	45615.000000	45615.000000
mean	106.932851	19.237444

	Character Count	Word Count
std	26.251517	4.935990
min	10.000000	1.000000
25%	89.000000	16.000000
50%	113.000000	20.000000
75%	128.000000	23.000000
max	200.000000	35.000000

6. **Sample Tweets:** We inspected sample tweets from each sentiment class to understand the language and characteristics of the dataset.

NEGATIVE Examples

- So disappointed in wwe summerslam! I want to see john cena wins his 16th title
- That sucks if you have to take the SATs tomorrow
- Amy Schumer sat down with The Hollywood Reporter and opened up about the sexist remarks and unnecessary criticism she's faced since

NEUTRAL Examples

- "Ben Smith / Smith (concussion) remains out of the lineup Thursday, Curtis #NHL #SJ"
- Sorry bout the stream last night I crashed out but will be on tonight for sure. Then back to Minecraft in pc tomorrow night.
- Chase Headley's RBI double in the 8th inning off David Price snapped a Yankees streak of 33 consecutive scoreless innings against Blue Jays

POSITIVE Examples

- "QT @user In the original draft of the 7th book, Remus Lupin survived the Battle of Hogwarts. #HappyBirthdayRemusLupin"
- @user Alciato: Bee will invest 150 million in January, another 200 in the Summer and plans to bring Messi by 2017"
- @user LIT MY MUM 'Kerry the louboutins I wonder how many Willam owns!!! Look Kerry Warner Wednesday!'

Initial EDA Findings

- TweetEval contains 59,899 tweets: 45,615 for training, 2,000 for validation and 12,284 for testing.
- Neutral is the largest class in all three dataset splits.
- Negative tweets make up about 15.6% of the training and validation sets, but 32.34% of the test set.
- Therefore, the class distribution of the test set is different from the training and validation sets.
- No missing text, empty tweets or missing labels were found.
- The training set contains 29 duplicate occurrences. Three duplicated text groups have conflicting labels.
- A training tweet contains about 19 words on average. Tweet lengths range from 1 to 35 words.
- We will use per-class precision, recall and F1-score because accuracy alone may hide weak performance for a specific sentiment class.
- Tweet length will also be used as metadata to check whether the models perform differently on short and long tweets.

Data Cleaning and Preparation

No missing text, empty tweets or missing labels were found.

The training set contains 29 duplicate occurrences, including three duplicated texts with conflicting labels. These records were retained to preserve the official TweetEval dataset, but the conflicts were noted as a data-quality limitation.

No extensive cleaning was required. Emojis, hashtags, punctuation, `@user`, `http` and negation words were retained because they may help sentiment classification.

The original text was preserved in a separate column, and any necessary preparation was applied to a processed-text column.

Baseline Models

We explore two baseline models to effectively perform performance evaluation for new candidate models and baseline models based on the "Ready to Ship" criteria and evaluations through metrics our team have selected.

In this section with the Provided Baseline Model Exploration. We use

1. **Twitter-roBERTa-base** for Sentiment Analysis: RoBERTa-base model trained with over 124 million tweets and finetuned with TweetEval benchmark.
2. **gpt2-finetuning-model-3000-samples**: gpt2 finetuned on the imdb dataset.
gpt2-finetuning-model-3000-samples model was originally trained for binary classification tasks.
Therefore, it does not fit in our multi-class classification situation. So we only use negative and positive without neutral to enable this model to perform.

Selected Candidate Models

We have selected two candidate models to conduct performance comparison with the baseline models.

The selected two candidate models are:

1. **BERTweet sentiment analysis ([link](#))**: This is a model based on BERTweet (a RoBERTa model trained on english tweets) and trained with SemEval 2017 corpus.
2. **Distilbert-base-sentiment ([link](#))**: This is a distilled model from the zero shot classification pipeline on the Multilingual Sentiment data.

Evaluation Metrics

To evaluate the candidate models and baseline models, we apply metrics as follows:

1. F1-score - Macro, Positive, Negative, Neutral
2. Accuracy
3. Precision - Positive, Negative, Neutral
4. Recall - Positive, Negative, Neutral
5. Confusion Matrix
6. Training speed

Purpose and use of each evaluation metrics

1. **F1-score:**

This metric is calculated by combining precision and recall. Thus, to achieve high F1-score, both precision and recall should be evenly high. We use Macro, Positive, Negative, and Neutral to inspect our model's

performance in different aspects. Macro F1 for overall, positive F1 for focused view on positive predictions, negative F1 for focused view on negative predictions, neutral F1 for focused view on neutral predictions.

2. **Accuracy:**

Accuracy is used to inspect how much the model predicted correctly among the whole data.

3. **Precision:**

Precision is calculated for positive, negative, and neutral. Precision refers to ratio of actual positives among all predicted positives. This applies the same to negative and neutral.

4. **Recall:**

Recall is also calculated for all positive, negative, and neutral. This indicates how many cases were predicted positive among actual positive data. This applies the same to negative and neutral.

5. **Confusion Matrix:**

Confusion matrix is a cross table indicating correct predictions and incorrect predictions. Confusion matrix will include: PREDICTED - positive, negative, neutral / ACTUAL - positive, negative, neutral - **Total: 6.**

6. **Prediction Speed:**

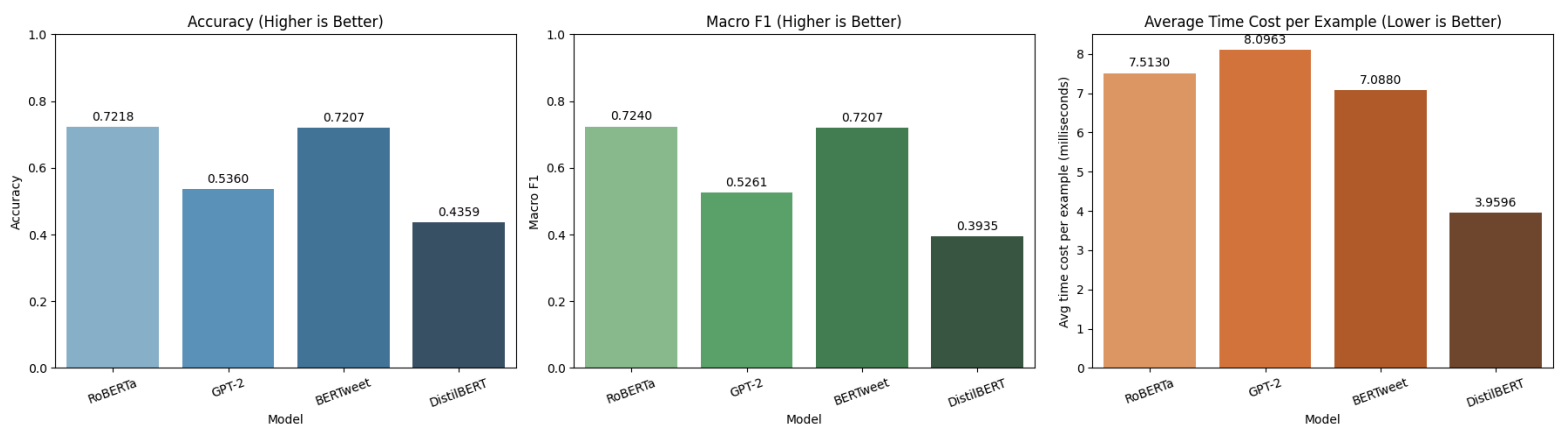
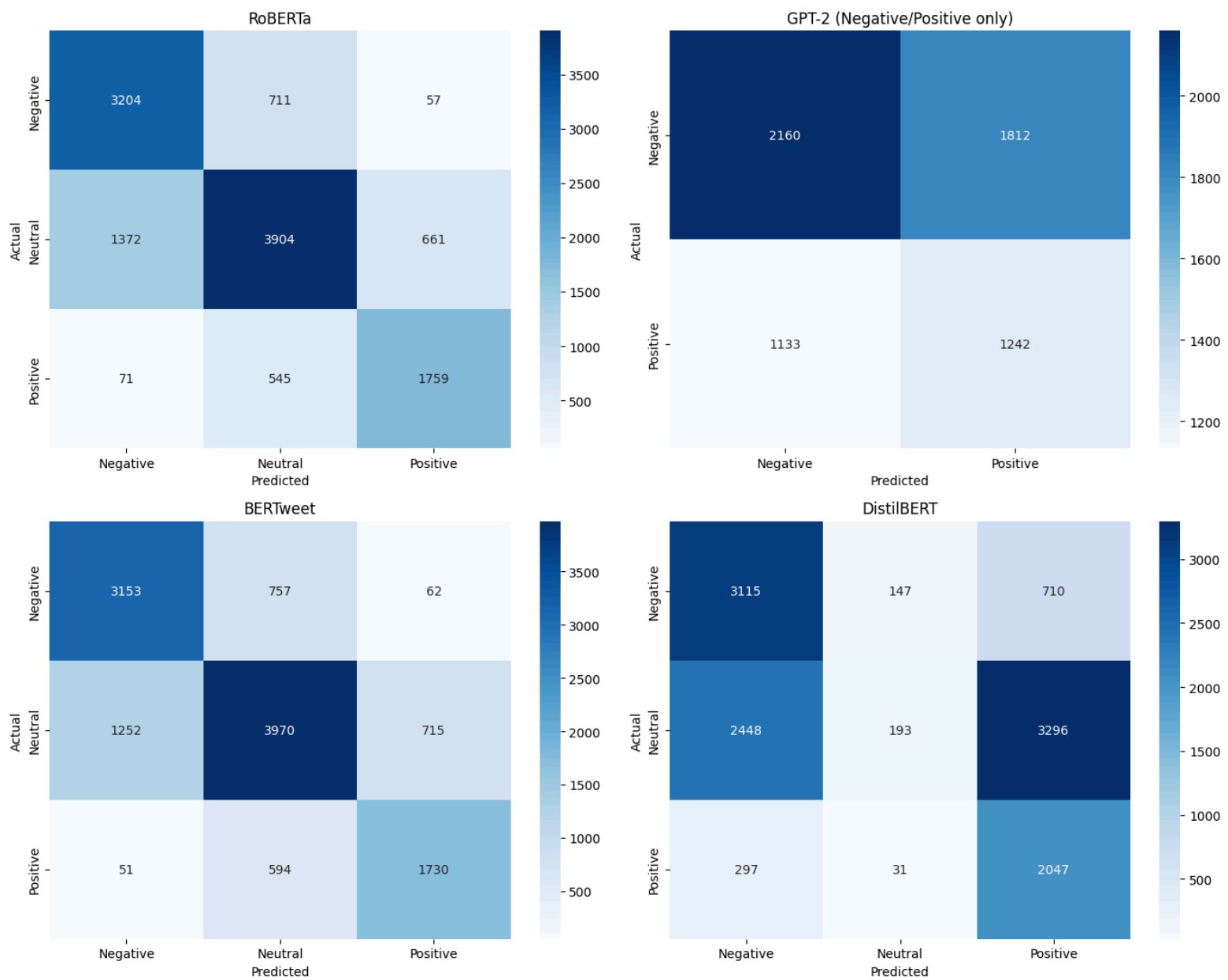
Number of posts and comments on the internet is rapidly rising. Fast prediction speed is important to follow the trend and ultimately to reflect user's needs and gain useful feedbacks from customers. By improving the prediction speed, models could be more suitable to follow rapidly changing environment. By evaluating the speed, we expect to inspect and compare which model is more suitable in the perspective of swiftness. To inspect the speed in various perspectives, we used Total prediction time (How long it took to handle for the whole data), Available examples per second (How many samples can be handled in a second), and Average time cost per example (How long it took to handle for one example - millisecond).

Initial Evaluation Result:

Model	Evaluation Scope	Number of Examples	Accuracy	Macro F1	Negative F1	Neutral F1
RoBERTa	Negative + Neutral + Positive	12284	0.7218	0.7240	0.7435	0.7036
GPT-2	Negative + Positive only	6347	0.5360	0.5261	0.5946	NaN
BERTweet	Negative + Neutral + Positive	12284	0.7207	0.7207	0.7482	0.7053
DistilBERT	Negative + Neutral + Positive	12284	0.4359	0.3935	0.6336	0.0612

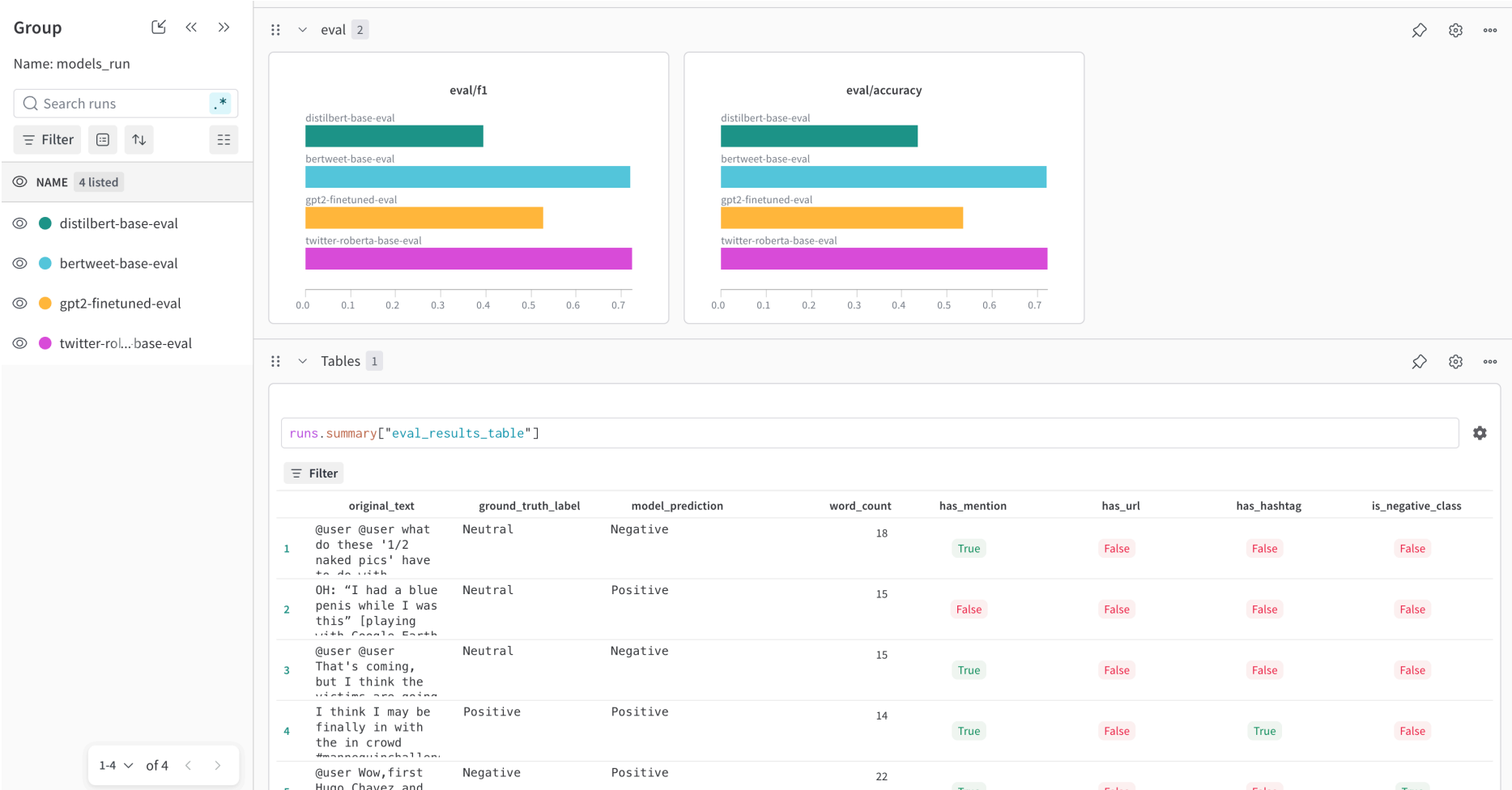
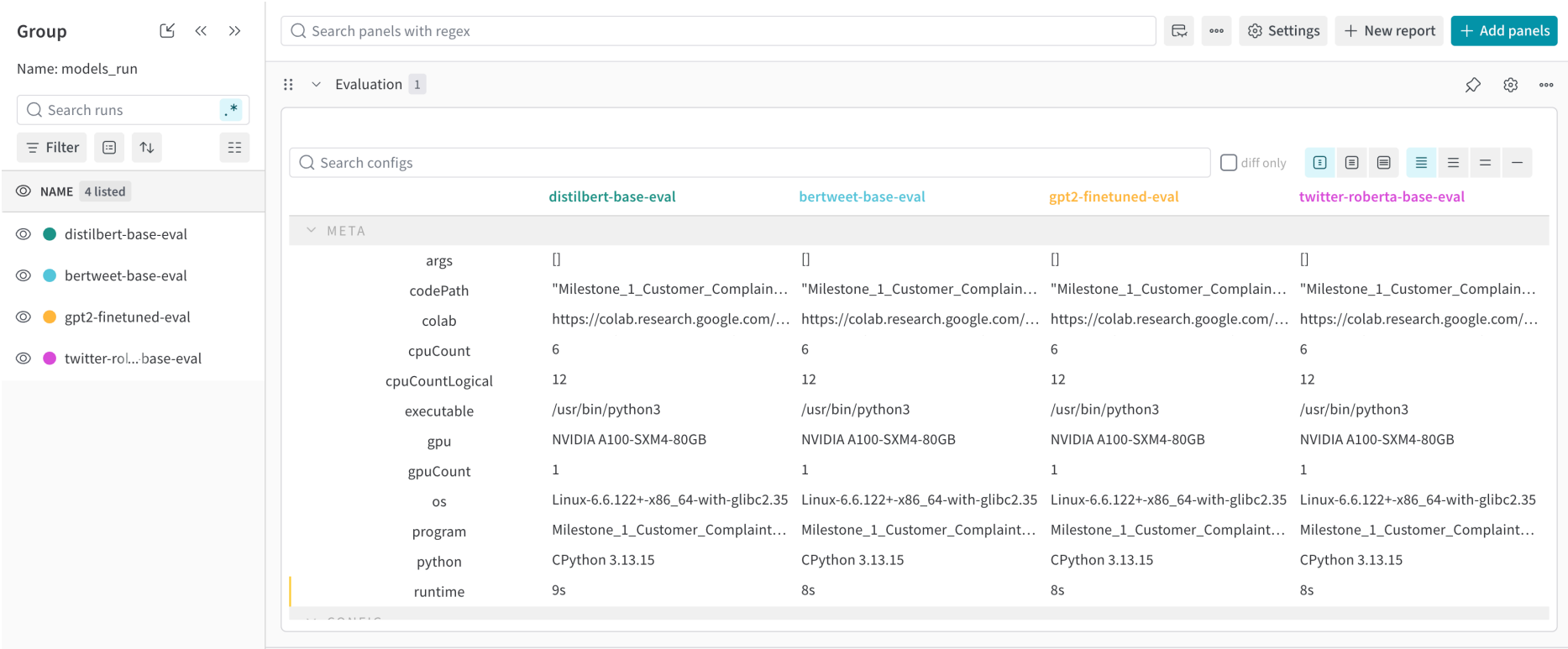
Model	Positive F1	Negative Precision	Neutral Precision	Positive Precision	Negative Recall	Neutral Recall
RoBERTa	0.7251	0.6895	0.7566	0.7101	0.8066	0.6576
GPT-2	0.4575	0.6559	NaN	0.4067	0.5438	NaN
BERTweet	0.7087	0.7076	0.7461	0.6901	0.7938	0.6687
DistilBERT	0.4858	0.5316	0.5202	0.3382	0.7842	0.0325

Model	Positive Recall	Total prediction time (seconds)	Available examples per second	Avg time cost per example (milliseconds)
RoBERTa	0.7406	27.2970	450.0121	2.2222
GPT-2	0.5229	13.1560	482.4406	2.0728
BERTweet	0.7284	25.0629	490.1278	2.0403
DistilBERT	0.8619	14.4663	849.1464	1.1777



Wandb

The code's execution & performance information has been added to wandb and data representations were conducted.



References

1. TweetEval Dataset: https://huggingface.co/datasets/cardiffnlp/tweet_eval
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4. Twitter-roBERTa-base for Sentiment Analysis: <https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>
5. GPT2-finetuning-sentiment-model-3000-samples: <https://huggingface.co/LYTinn/gpt2-finetuning-sentiment-model-3000-samples#gpt2-finetuning-sentiment-model-3000-samples>
6. BERTweet sentiment analysis: <https://huggingface.co/finiteautomata/bertweet-base-sentiment-analysis>
7. Distilbert-base-sentiment (multilingual-case): <https://huggingface.co/lxyuan/distilbert-base-multilingual-cased-sentiments-student>
8. <https://www.cxtoday.com/uncategorized/the-importance-of-customer-complaints/>